

Stable Lignin-Rich Nanofibers for Binder-Free Carbon Electrodes in Supercapacitors

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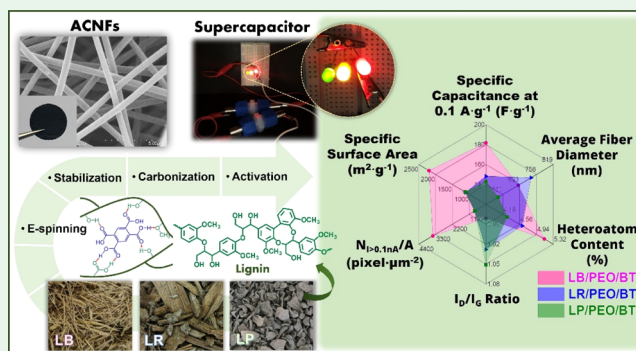
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Supporting Information

ABSTRACT: In this work, 1,2,4,5-benzenetetracarboxylic acid (BT) is used as a cross-linker to improve the oxidative thermostabilization of lignin-rich nanofibers so that the activated carbon nanofiber (ACNF) sheets can be prepared from lignins. As sources, we used organosolv lignins from sugarcane bagasse (LB), para rubber wood chip (LR), and palm kernel shell (LP). The addition of BT imparts hydrogen bond interactions between BT and lignin and increases the oxidation degree of lignin during the stabilization, enabling retention of nanofiber morphology after heat treatments. The preparation of activated carbon in the form of nanofibers substantially enhances the specific surface area of the materials when compared with the activated carbon powders from pure lignins. The properties of the resulting ACNFs are influenced by the structure and composition of lignins. With the synergistic effects of a small fiber diameter (688 ± 80 nm), a high content of heteroatom (O atomic content $\sim 5.1\%$), a high graphitic carbon structure (I_D/I_G ratio ~ 0.98), a high conductivity ($N_{I>0.1\text{ nA}}/A \sim 3479 \pm 404$ pixel μm^{-2} at 0.5 V), and a high specific surface area (~ 2195 $\text{m}^2 \text{g}^{-1}$), the ACNFs from LB show the highest specific capacitance of 182 F g^{-1} (at a current density of 0.1 A g^{-1} in $1.0 \text{ M H}_2\text{SO}_4$ electrolyte) with good cycling stability (95.7% capacitance retention after 6000 cycles). A maximum energy density of 31 W h kg^{-1} at a power density of 240 W kg^{-1} can be achieved from the LB/PEO/BT ACNFs. Their performance is almost comparable to that of polyacrylonitrile ACNFs. These findings demonstrate a simple and effective approach to produce lignin-based ACNFs from diverse biomass feedstock for sustainable development of renewable materials for supercapacitor electrodes.

KEYWORDS: lignin, activated carbon nanofiber, crosslink, thermostabilization, carbon electrode, supercapacitor



INTRODUCTION

Electrochemical devices are one of the crucial technologies for energy storage. Among different types of electrochemical energy storage devices, supercapacitors have gained attention for an efficient delivery of electrical energy due to their high power density, fast charge/discharge capability, superior cycling stability, and safe operation.^{1,2} The working principle of electrochemical double-layer capacitors (EDLCs) is based on charge accumulation at electrode/electrolyte interfaces. Thus, the efficiency of supercapacitors depends significantly on properties of the electrode materials. The key requirements of electrode materials are high specific surface area with an appropriate pore structure,^{3–5} good electrical conductivity,^{6–8} and electrochemical stability.^{9,10} Numerous carbonaceous materials, such as activated carbon,^{8,11} carbon nanotube,¹² graphene,^{13,14} graphene quantum dots,¹⁵ carbon aerogel,^{16,17} and carbon nanofibers,^{17,18} have been used as active materials for electrodes. Mostly, the active materials and other additives in the form of powder require a binder to hold the materials in the electrodes and on the collectors.^{19–21} However, many

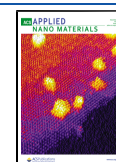
commonly used binders, such as polytetrafluoroethylene and poly(vinylidene fluoride), have high electrical resistivity.^{22–24} Therefore, the performance of the electrodes is lowered owing to a reduction in accessible surface area for ion adsorption and reduced electrical conductivity.

Among various types of active materials, porous carbon nanofiber sheets offer a great potential for binder-free electrodes because of their high specific surface area and porosity, good electrical conductivity, and flexibility.^{25–28} Porous carbon nanofiber sheets are prepared in two main steps: nanofiber fabrication of carbon precursors via electrospinning and heat treatments including stabilization, carbonization, and activation. Polyacrylonitrile (PAN) is the most

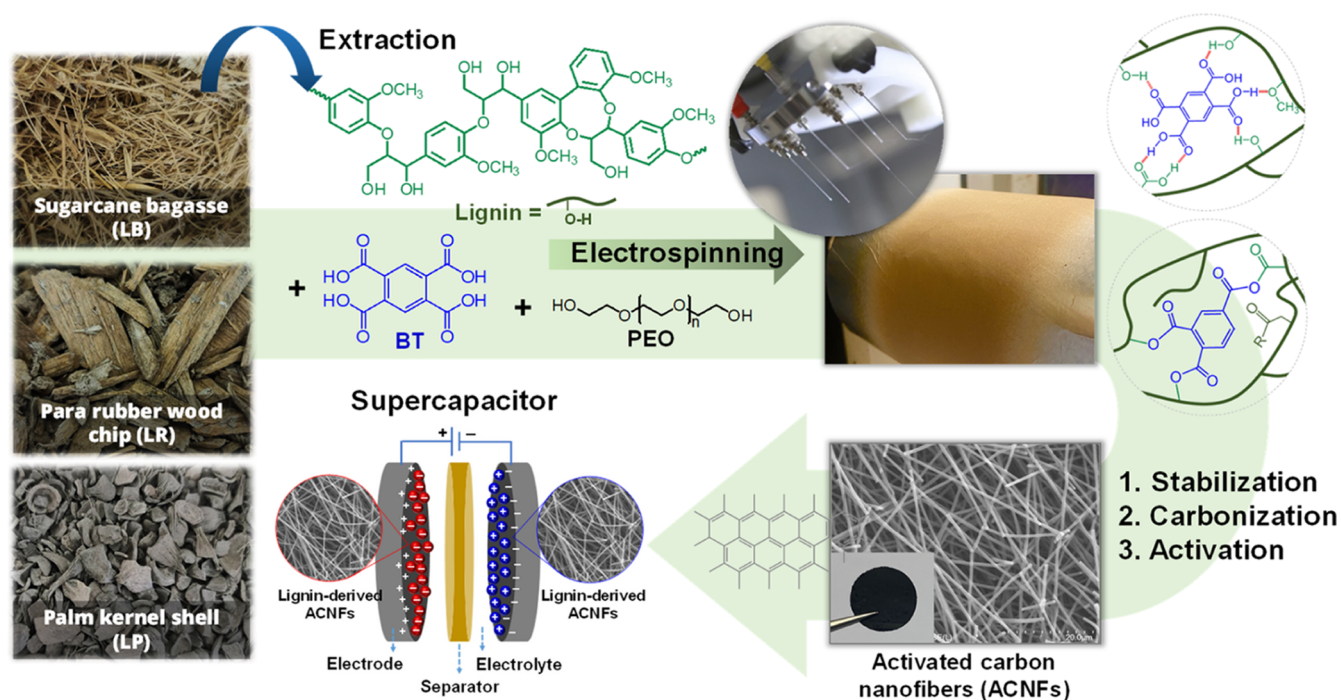
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Scheme 1. Schematic Illustration of the Preparation Process of ACFN Sheets from Stable Lignin-Rich Nanofibers for Binder-Free Carbon Electrodes in Supercapacitors



used precursor for carbon fibers due to its high carbon yield, excellent spinnability, and good mechanical properties.²⁹ However, PAN is from fossil fuel resources and expensive, limiting its broad applications in energy storage.^{30,31}

With growing demand of ecofriendly and renewable materials, lignocellulosic biomass is of great interest as a biobased precursor with low cost for carbon fibers. In fact, lignocellulosic biomass can be directly used as a source for carbon materials.^{32–34} However, cellulose has been widely utilized in various industries such as paper, textile, cosmetic, and pharmaceutical industries, whereas lignin is considered as waste and still underutilized. Due to its aromatic structure with high carbon content and availability, lignin is a good candidate precursor for producing carbon fibers. Lignin is composed of three main units: syringyl (S), guaiacyl (G), and *p*-hydroxyphenyl (H). The structure of lignin depends on many factors such as the plant, source, extraction procedure, and environment.³⁵ This variability leads to the variation in properties of the resulting carbon materials.^{36–39} Although a number of methods to produce activated carbon materials from lignins have been reported for supercapacitor electrodes,^{40,41} the conversion of lignin nanofibers into carbon fibers while preserving their original shape remains challenging.

Thermostabilization is a key process to prevent fusion of the material structure prior to subsequent carbonization. Thermostabilization involves heating of the precursor in air in a temperature range of 200–300 °C.⁴² During thermostabilization, thermosetting properties are only achieved by providing sufficient oxygen for changing the structure of lignin by bond cleavage and oxidation including dehydration, cyclization, and cross-linking.^{35,43,44} In some cases, the thermostabilization process required from ~7 h to more than 3 days depending on the types of lignin and fiber compositions to achieve infusible fibers.^{31,45–47} Therefore, various alternative approaches have been studied to improve the stabilization of lignin fibers. Yue et

al. reported that short-time stabilization of organosolv switchgrass lignin and hardwood lignin fibers via a two-step heating process resulted in fused fiber bundles.⁴⁸ After treating with hydrochloric acid (HCl) gas in air and further stabilization, the resulting fibers were infusible. Zhang et al. successfully stabilized lignin fibers without fusion by a UV treatment.⁴⁹ The FTIR results indicated that UV treatment increased the formation of carbonyl (C=O) groups and led to degradation of aromatic rings. However, the above two methods required an additional step of either HCl or UV pretreatment before thermostabilization.

To our idea, 1,2,4,5-benzenetetracarboxylic acid (BT), an aromatic ring containing multicarboxylic acid moieties, is a possible molecule to improve the thermostabilization of the lignin-rich nanofibers via hydrogen bond interactions and condensations between lignin and BT. Thus, in this study, we use BT as a cross-linker to stabilize lignin-rich nanofibers. By combining lignin with minor poly(ethylene oxide) (PEO) in the presence of BT, activated carbon nanofiber (ACNF) sheets without fiber deformation can be achieved from a variety of biomass feedstock (Scheme 1). The in situ cross-linking of lignin by BT during thermostabilization makes this approach simple and useful for the fabrication of ACFNs as binder-free electrodes in supercapacitors.

MATERIALS AND METHODS

Materials. Lignins from three different biomass feedstock (Table S1), that is, sugarcane bagasse (LB), para rubber wood chip (LR), and palm kernel shell (LP), were used as starting materials. LB represents herbaceous biomass, while LR and LP are residues of dicot and monocot plants, respectively. Sugarcane bagasse was obtained from Mitr Phol Sugar Corp., Ltd., Thailand. Para rubber wood chip and palm kernel shell were provided by Asia Biomass Co., Ltd., Thailand. The lignins were extracted via the organosolv process using 70% aqueous ethanol and precipitated by addition of water to the liquid fractions. The extracted LB, LR, and LP powders were dried in a hot

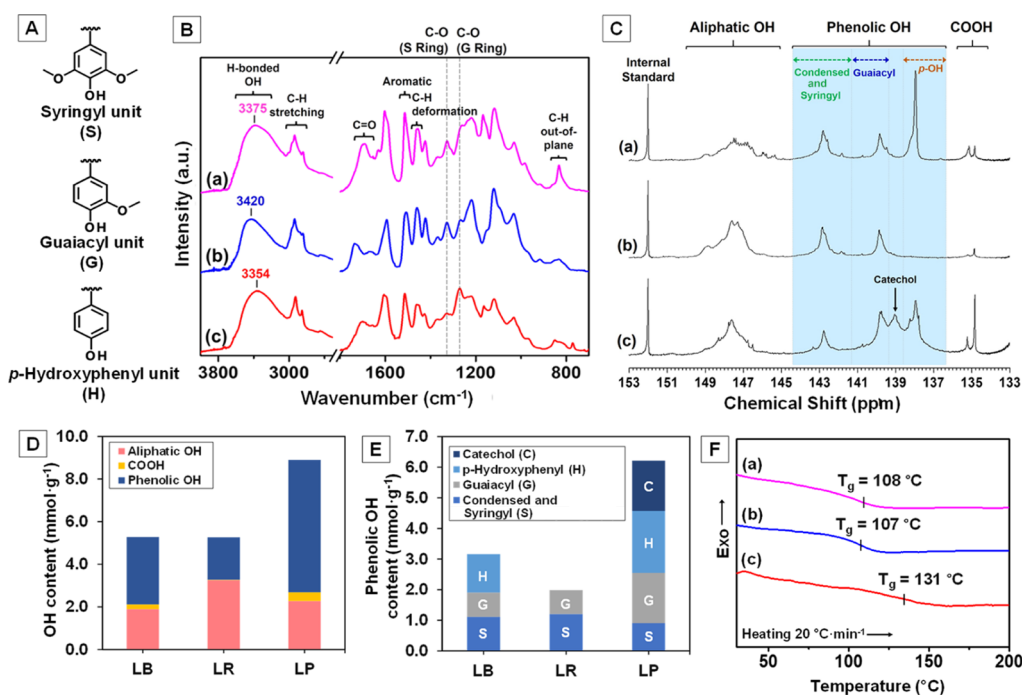


Figure 1. (A) Three major aromatic units of lignins, (B) ATR–FTIR spectra, (C) ^{31}P NMR spectra, (D) plot of OH content, and (E) plot of phenolic OH content calculated from ^{31}P NMR analysis, and (F) DSC thermograms of (a) LB, (b) LR, and (c) LP.

air oven at 70 °C for 4 h before preparing electrospinning solutions. PEO ($M_w = 600 \text{ kg mol}^{-1}$), BT, and PAN ($M_w = 150 \text{ kg mol}^{-1}$) were purchased from Sigma-Aldrich. N,N -Dimethylformamide (DMF) was purchased from Carlo Erba. All chemicals were used as received without further purification.

Structural Characterization of Lignin. Attenuated total reflectance–Fourier transform infrared (ATR–FTIR) spectra were obtained using a Thermo Nicolet iS 5 FTIR spectrometer. The ATR–FTIR data were collected at 128 scans with a resolution of 2 cm^{-1} . The quantitative 2D heteronuclear single quantum coherence (2D-HSQC) nuclear magnetic resonance (NMR) and ^{31}P NMR measurements were conducted on a Bruker Ultra Shield AV-500. 2D-HSQC NMR was used to determine the S/G ratio of lignins. For the 2D-HSQC measurement, dried lignin (50 mg) was dissolved in $\text{DMSO}-d_6$ (0.5 mL), and the experiment was performed according to the previous method with minor modifications.⁵⁰ The 2D-NMR spectra were processed using the MestReNova program, and the S/G ratio was calculated as reported by Wang et al.⁵¹ ^{31}P NMR was used to quantitatively determine the amount and distribution of hydroxyl groups in lignin isolated from different biomass feedstock. Sample preparation for ^{31}P NMR analysis was carried out as reported elsewhere.⁵² Basically, lignin was phosphorylated with 2-chloro-4,4,5,5-tetramethyl-1,3,2-dioxaphospholane (TMDP) to differentiate different types of hydroxyl groups, which are aliphatic OH, carboxyl OH (COOH), and phenolic OH including syringyl (S), guaiacyl (G), and *p*-hydroxyphenyl (H) units (Figure 1A). In brief, lignin powder was weighed (13.4–14.2 mg) and dissolved in DMF (79 μL). The resulting brown solution was diluted with deuterated chloroform solution (158 μL), followed by the serial addition of pyridine (53 μL), *N*-hydroxy-5-norbornene-2,3-dicarboximide (4.5 mg mL^{-1}) in 1.6:1.0 v/v pyridine/ CDCl_3 (105 μL), chromium (III) acetylacetonate (15.8 mg mL^{-1}) in 1.6:1.0 v/v pyridine/ CDCl_3 (26 μL), and TMDP (79 μL). The resulting mixture was vigorously mixed using a vortex mixer. Chromium (III) acetylacetonate was used as a relaxation reagent in ^{31}P NMR analysis. The amount of different OH groups was evaluated from the integral ratios using *N*-hydroxy-5-norbornene-2,3-dicarboximide as an internal standard.

Preparation of Lignin-Rich Nanofibers by Electrospinning. The electrospinning solutions were prepared by mixing LB, LR, and LP powders with PEO in DMF at 45 wt % with a weight ratio of

lignin/PEO of 99:1. BT was added to the mixed solutions at 10 wt % of lignin/PEO and mixed by stirring at 80 °C for 1 h. After cooling, the mixed solutions were loaded into a syringe and extruded at a constant flow rate of 0.1 mL h^{-1} per needle. The dc power was applied between a needle and a rotating collector at a voltage of 7 kV. The tip-to-collector distance was 15 cm. The electrospinning was done at ambient temperature with a take-up speed of the collector of 100 rpm. The nanofibers containing LB, LR, and LP with 1 wt % PEO are named as LB/PEO, LR/PEO, and LP/PEO, respectively. The LB/PEO, LR/PEO, and LP/PEO nanofibers with 10 wt % BT are denoted as LB/PEO/BT, LR/PEO/BT, and LP/PEO/BT, respectively.

Conversion of Lignin Nanofibers to ACFNs by Heat Treatments. Three methods were used to prepare ACFNs: (i) stabilization in air by heating from 30 °C to 250 °C at a heating rate of 0.5 °C min^{-1} and holding at 250 °C for 1 h; (ii) carbonization by heating from 30 °C to 900 °C at a heating rate of 10 °C min^{-1} and holding at 900 °C for 2 h under N_2 flow (100 L h^{-1}); (iii) activation by raising the temperature from 30 °C to 900 °C at a heating rate of 10 °C min^{-1} and keeping isothermally at 900 °C for 15 min under N_2 flow (100 L h^{-1}) and for 1 h under CO_2 flow (100 L h^{-1}).

Materials Characterization. Differential scanning calorimetry (DSC) thermograms were obtained using a NETZSCH DSC 214 Polyma. The DSC analysis was performed within the temperature range of 0–200 °C with a heating–cooling–heating scan at a heating rate of 20 °C min^{-1} under a nitrogen atmosphere. The glass transition temperatures of the pristine lignins and nanofibers were evaluated from the second heating scan. Thermogravimetric analysis (TGA) was carried out on a Mettler Toledo STARE SYSTEM TGA/DSC 3+ at a heating rate of 10 °C min^{-1} and a flow rate of N_2 40 mL min^{-1} . Scanning electron microscopy (SEM) images were obtained using a Hitachi S-3400N. For SEM, the samples were sputtered with Au at 15 mA for 90 s. Transmission electron microscopy (TEM) images were acquired on a JEOL JEM-2100Plus electron microscope operating at 200 kV. X-ray diffraction (XRD) patterns were collected on a Bruker D8 ADVANCE diffractometer with $\text{Cu K}\alpha$ radiation. The XRD measurement was carried out at a voltage of 40 kV and a current of 40 mA with a resolution of $0.02^\circ 2\theta$ and 0.5 s per step. Surface elemental compositions were analyzed by using X-ray photoelectron spectroscopy (XPS, Shimadzu AXIS Supra+). Raman spectra were recorded

Table 1. S/G Ratios Estimated from 2D-HSQC NMR Results and Hydroxyl Group Contents of Lignins from Different Sources Quantified by ^{31}P NMR Analysis

sample	S/G ratio	hydroxyl group content						
		aliphatic OH (mmol g $^{-1}$)	COOH (mmol g $^{-1}$)	phenolic OH (mmol g $^{-1}$)				total OH (mmol g $^{-1}$)
				condensed and syringyl	guaiacyl and catechol	p-OH	total	
LB	3.6	1.90	0.22	1.12	0.79	1.25	3.16	5.28
LR	5.7	3.26	0.02	1.21	0.78	0.00	1.99	5.27
LP	0.5	2.28	0.42	0.91	3.27	2.03	6.21	8.91

from 3200 to 870 cm^{-1} by using a NT-MDT NTEGRA Spectra system with a green laser ($\lambda = 532 \text{ nm}$). N_2 adsorption–desorption isotherms were analyzed on a Micromeritics 3Flex physisorption instrument at 77 K. The samples were degassed under vacuum at 250 $^\circ\text{C}$ for 16 h before Brunauer–Emmett–Teller (BET) surface area analysis.

Conductive atomic force microscopy (cAFM) measurement was carried out on a JPK NanoWizard 3 in Quantitative Imaging (QI) mode.^{53,54} The ACNFs were dispersed in ethanol. 10 μL of the solutions was drop-cast on a 500 Å thick gold film that was evaporated on a diced silicon wafer. To ensure good electrical contact, the gold substrates were cleaned in oxygen plasma before sample casting. The samples were dried at 120 $^\circ\text{C}$ under vacuum for 6 h prior to cAFM scanning. The cAFM images were recorded using a PtIr-coated electrically conductive probe (SCM-PIT-V2, Bruker). The cAFM scanning parameters were set as follows: set point 25 nN, Z-length 2000 nm, pixel time 20 ms, scan size $4 \times 4 \mu\text{m}^2$, and mapping resolution 256×256 pixels. The cAFM data analysis was carried out using Gwyddion (version 2.55).

Electrochemical Characterization. Electrochemical performance of the ACNF sheets including cyclic voltammetry (CV), galvanostatic charge/discharge (GCD), and electrochemical impedance spectroscopy (EIS) was measured on an Autolab PGSTAT12 potentiostat/galvanostat electrochemical system using two symmetrical electrode cells. The ACNF sheets were cut into a circular shape with a diameter of 5 mm and assembled into the cell using Whatman no.1 paper as a separator. H_2SO_4 solution (1.0 M) was used as an electrolyte. The CV measurement was conducted at different scan rates, that is, 5, 10, 20, 50, 100, and 200 mV s^{-1} within a voltage range of -0.45 to $+0.65 \text{ V}$. The GCD profiles were measured at various current densities ranging from 0.1 to 5.0 A g^{-1} . EIS analysis was collected in a frequency range from 100 kHz to 0.1 Hz at an AC amplitude of 10 mV. The specific capacitance of the two-electrode system was calculated from the GCD curves using eq 1

$$C_s = \frac{4I \times \Delta t}{m \times \Delta V} \quad (1)$$

where C_s is the gravimetric specific capacitance (F g^{-1}), I is the discharge current (A), Δt is the discharge time (s), m is the total mass of the two-electrode materials (g), and ΔV is the discharge voltage range (V).

The energy density and power density of the materials were estimated by using

$$E = \frac{C_s \times \Delta V^2}{2 \times 3.6} \quad (2)$$

$$P = \frac{E \times 3600}{\Delta t} \quad (3)$$

where E is the energy density (W h kg^{-1}) and P is the power density (W kg^{-1}).

RESULTS AND DISCUSSION

Compositions of Lignins Extracted from Different Biomass Feedstock and Their Thermal Properties. Figure 1B displays ATR–FTIR spectra of the pristine LB, LR, and LP. The FTIR spectra of LB, LR, and LB show typical

absorption bands of lignin at ~ 3600 – 3200 cm^{-1} (H-bonded OH stretching in phenolic and aliphatic structures), ~ 2940 – 2840 cm^{-1} (asymmetric and symmetric C–H stretching in methyl and methylene groups), ~ 1730 – 1694 cm^{-1} ($\text{C}=\text{O}$ stretching), ~ 1513 – 1508 cm^{-1} (aromatic skeletal vibration), ~ 1458 – 1456 cm^{-1} (C–H deformation in methyl and methylene groups), and ~ 850 – 830 cm^{-1} (C–H out-of-plane deformation of aromatic rings).^{55,56} The OH stretching band of LP and LB is broader than that of LR. As revealed by Kubo et al.,⁵⁵ the broader the OH stretching band, the more extensive the hydrogen bonds due to various types of H-bonded OH presenting in lignin. The H-bonded OH stretching band of LP is centered at 3354 cm^{-1} while that of LB and LR shifts to higher wavenumbers at 3375 and 3420 cm^{-1} , respectively. This suggested that the hydrogen bond interactions of LP were potentially stronger than those of LB and LR, respectively.

To compare relative monomer compositions of each lignin, syringyl/guaiacyl (S/G) ratios were estimated from the FTIR and NMR data. For the FTIR spectra, the absorption bands at ~ 1327 – 1326 and ~ 1271 – 1258 cm^{-1} are assigned to C–O of the S ring and G ring, respectively. The ratios of the C–O (S-ring) peak intensity at ~ 1327 – 1326 cm^{-1} relative to the C–O (G-ring) peak intensity at ~ 1271 – 1258 cm^{-1} were 0.77, 0.95, and 0.62 for LB, LR, and LP, respectively. For the quantitative 2D-HSQC NMR analysis (Figure S1 in the Supporting Information), the S/G ratios of the pristine LB, LR, and LP are listed in Table 1. Similar tendency was observed for the calculation based on 2D-HSQC NMR and FTIR, that is, the S/G ratios were in the order of $\text{LR} > \text{LB} > \text{LP}$.

Figure 1C shows ^{31}P NMR spectra of LB, LR, and LP with peak assignments of various hydroxyl groups.^{52,57} Lignin is a complex phenolic polymer consisting of various types of hydroxyl groups including aliphatic OH, COOH, and phenolic OH. After phosphorylation of OH groups with TMDP, the amount of hydroxyl groups could be calculated from the integrals of different OH groups in lignins divided by the integral of the internal standard with accurate concentration in relation to the mass of lignin powders. The contents of OH groups calculated from ^{31}P NMR analysis are summarized in Table 1. The distribution of OH groups in each lignin is plotted in Figure 1D. LP has a higher number of hydroxyl groups than LB and LR. The phenolic OH is the major component in LP (phenolic OH $\sim 70\%$ of total OH) and LB (phenolic OH $\sim 60\%$ of total OH), whereas the aliphatic OH is the main component in LR (aliphatic OH $\sim 62\%$ of total OH). A minor amount of the COOH group (0.3–4.7% of total OH) is found in lignins. The COOH content of LP is greater than that of LB and LR. Comparing the compositions of phenolic OH (Figure 1E), it can be found that LB is composed of S, G, and H units, whereas LR is composed of only S and G units. LP contains 14.6% condensed and S units, 26.4% G unit, and 32.7% H unit including 26.3% catechol which is a

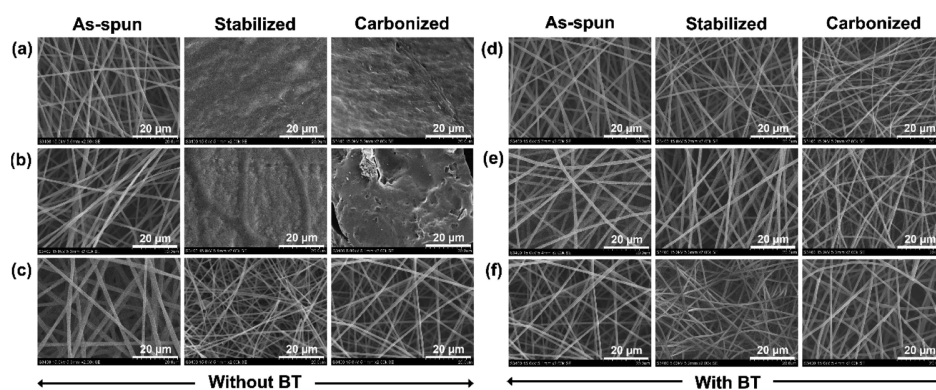


Figure 2. SEM images of (a) LB/PEO, (b) LR/PEO, (c) LP/PEO, (d) LB/PEO/BT, (e) LR/PEO/BT, and (f) LP/PEO/BT nanofibers before and after stabilization and carbonization.

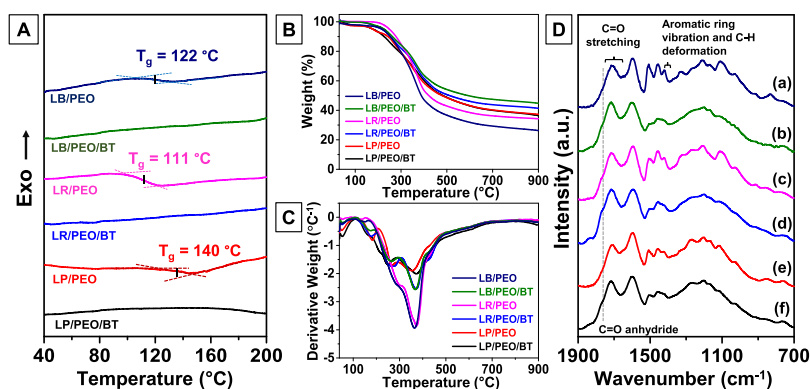


Figure 3. (A) DSC, (B) TG, and (C) DTA thermograms of the as-spun lignin-rich nanofibers and (D) FTIR spectra of (a) LB/PEO, (b) LB/PEO/BT, (c) LR/PEO, (d) LR/PEO/BT, (e) LP/PEO, and (f) LP/PEO/BT nanofibers after stabilization at 250 °C in air.

demethylated unit of guaiacol. It was reported that the S/G content of lignin had an influence on the properties of carbon materials.^{36,38} Based on ³¹P NMR calculation, the S/G ratios of LB, LR, and LP were 1.4, 1.6, and 0.6, respectively. Consistent with the FTIR and 2D-HSQC NMR findings, the S/G ratio of LR was greater than that of LB and LP, respectively.

The DSC thermograms of the pristine lignin powders are displayed in Figure 1F. The inflection points are observed in the temperature range of 107–131 °C, reflecting the glass transition temperature (T_g) of lignin. The T_g of LP (131 °C) is higher than that of LB (108 °C) and LR (107 °C), respectively. The DSC results were in good agreement with the findings from FTIR and NMR analyses. The stronger hydrogen bond network accompanied by the lower S/G ratio of LP, compared with the others, diminished the thermal mobility of lignin, resulting in higher T_g . This is due to the fact that the G unit with one methoxyl group has less free volume than the S unit with two methoxyl groups. The lower the free volume, the lower the thermal mobility.⁵⁶

Activated Carbon Powders from Lignins. For comparison, the organosolv lignin powders were heat-treated by using the same thermal treatment processes (stabilization, carbonization, and activation) as for the preparation of ACNFs. Figure S2 shows SEM images of the pristine lignin powders before and after stabilization, carbonization, and activation. After heat treatments, LB and LR particles fused, resulting in a homogeneous mass of carbon with a smooth surface. In contrast, LP-derived activated carbon appeared as small particles with a highly rough surface, similar to the precursor. This might be due to the strong intermolecular hydrogen bond

interactions of LP, as proven by FTIR. As revealed by Kubo and Kadla, hydrogen bonds had a vital role in thermal stabilization of lignin.⁵⁵ However, they found that the aliphatic OH of lignin formed stronger and more complex hydrogen bonds than the phenolic OH. In our case, although LP contained lower aliphatic hydroxyl groups than LR, its hydrogen bonds were stronger, resulting in the retention of morphology of LP powder after thermal stabilization, carbonization, and activation. In contrast, the fusible structures of LB and LR with high S/G ratios were due to their weak intermolecular hydrogen bonds and high thermal mobility (low T_g).

The porosity of the activated carbon powders from different lignins was characterized by N₂ adsorption–desorption isotherms at 77 K. All the activated carbon powders present a steep increment of nitrogen uptake at low relative pressure (P/P_0), revealing a type I isotherm (Figure S3a). This indicated the presence of microporous structures. The pore size distribution in Figure S3b reveals that the pore size of all samples is identical, that is, ~0.5–0.8 nm. Other textural properties of the lignin-derived activated carbon powders are listed in Table S2. The specific surface area of the activated carbon powders from LB, LR, and LP was 590, 534, and 453 m² g^{−1}, respectively. Their total pore volume was in the range of 0.17–0.23 cm³ g^{−1} with a micropore volume of ~95–100% and a mesopore volume of ~0–5%. It should be noted that although LP exhibited separate small particles with a rough surface after activation, its surface area and total pore volume were lower than those of LB and LR.

Electrospinning and Conversion of Lignin-Rich Nanofibers into Carbon Nanofibers. DMF is a good solvent for organosolv lignin; therefore, it was used for preparing lignin solutions for electrospinning. Because pure lignin solutions in DMF at various concentrations were not spinnable, PEO as a fiber-forming polymer was mixed to improve the spinnability of lignin. Lignin/PEO blend nanofibers were successfully prepared at different lignin/PEO ratios. 1 wt % of PEO in the blends was the minimum content to allow fiber generation. Therefore, lignin/PEO solutions with a weight ratio of lignin/PEO for 99:1 were chosen for preparing all electrospun nanofibers. Dry weight contents of the solutions were varied from 30 to 50 wt %. Bead-free and continuous nanofibers could be obtained from the solutions containing 45 wt % dry weight contents of lignin/PEO (99:1). Figure 2a–c displays the SEM images of LB/PEO, LR/PEO, and LP/PEO nanofibers before and after stabilization and carbonization. Similar to the lignin powders, after being stabilized at high temperature, LB/PEO and LR/PEO nanofibers are fused together, while fibrous structures of LP/PEO are preserved. The carbon fibers obtained from LP/PEO are smooth and uniform. The destroyed structures of LB/PEO and LR/PEO nanofibers were due to the higher S/G ratio and weaker hydrogen bonds of LB and LR compared to LP, resulting in low T_g and fusion of LB/PEO and LR/PEO nanofibers. As observed in the DSC thermograms (Figure 3A), T_g s of lignins in LB/PEO, LR/PEO, and LP/PEO nanofibers appear at 122, 111, and 140 °C, respectively.

The LB/PEO/BT, LR/PEO/BT, and LP/PEO/BT nanofibers were successfully prepared by electrospinning. As demonstrated in the SEM images (Figure 2d–f), the nanofibrous structures of LB/PEO/BT, LR/PEO/BT, and LP/PEO/BT are maintained after being subjected to thermostabilization at 250 °C. After carbonization, the lignin nanofiber sheets obtained dramatically shrank, but their fibrous morphology under SEM observation remained substantially unchanged. When combining with BT, the T_g s of lignins were not detected up to 200 °C (Figure 3A). This result verified that BT acted as a cross-linker to prevent thermal mobility or softening of lignin and thus hindered the fusion of the nanofibers during stabilization.

TGA was measured to study the effect of BT on the thermal stability of lignins. There are three main degradation steps in the temperature ranges of 50–110, 120–300, and >300 °C (Figure 3B,C). The first weight loss at a temperature less than 110 °C was attributed to the evaporation of absorbed moisture. The second weight losses at 120–300 °C were associated with cleavage of α -/ β -aryl-alkyl-ether linkages, dehydration of aliphatic OH, condensation, and decarboxylation.^{58,59} In this region, DTA peaks at 120–230 °C were clearly observed only in LP/PEO and all lignin/PEO with BT. We speculated that this might be due to the presence of COOH in lignin and BT, accelerating cross-linking of lignins during oxidative thermostabilization. However, the decomposition temperature of lignins was slightly reduced when adding BT. The decomposition at 300–600 °C was due to degradation of aromatics and C–C bonds.

To clarify the influence of BT on the thermostabilization of lignin, FTIR analysis of the lignin/PEO nanofibers both with and without BT after being stabilized at 250 °C was investigated. The adsorption band at 1716–1709 cm^{-1} corresponds to unconjugated C=O stretching in ketones, carbonyls, and ester groups, and the adsorption band at 1431–

1423 cm^{-1} corresponds to the aromatic ring vibration combined with C–H in-plane deformation (Figure 3D).^{43,60,61} The intensity ratio of C=O stretching at 1716–1709 cm^{-1} and aromatic ring vibration/C–H deformation at 1431–1423 cm^{-1} ($I_{\text{C=O}}/I_{\text{aromatic/C-H}}$) was estimated. It was revealed that the peak of the aromatic ring vibration and C–H in-plane deformation at $\sim 1425 \text{ cm}^{-1}$ was insignificantly affected by stabilization at 250 °C.^{44,60} It was used as a reference peak to compare the relative degree of oxidation from the $I_{\text{C=O}}/I_{\text{aromatic/C-H}}$ ratios of the fibers after stabilization.⁶⁰ The $I_{\text{C=O}}/I_{\text{aromatic/C-H}}$ ratios of the stabilized LB/PEO, LR/PEO, and LP/PEO were 1.09, 1.10, and 1.18, respectively. Evidently, the addition of BT for 10 wt % in the LB/PEO, LR/PEO, and LP/PEO led to an increase of $I_{\text{C=O}}/I_{\text{aromatic/C-H}}$ ratios up to 1.51, 1.19, and 1.33, respectively. This result confirmed the contribution of BT to enhance the formation of unconjugated C=O bonds and improve the oxidation degree of lignin during stabilization. In addition, the appearance of the peak at $\sim 1770 \text{ cm}^{-1}$ of the lignin/PEO/BT indicated the formation of C=O anhydrides by condensation between COOH groups of lignin and BT.

Characterization of ACNFs. SEM and TEM were used to investigate the morphologies and microstructures of the ACNFs from different lignins. From the SEM images (Figure 4A), all ACNFs are relatively uniform and smooth. The

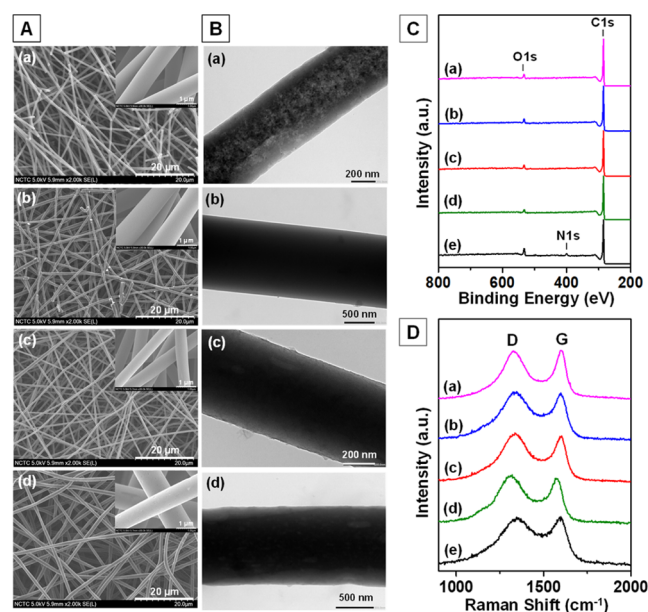


Figure 4. (A) SEM images, (B) TEM images, (C) XPS survey spectra, and (D) Raman spectra of (a) LB/PEO/BT, (b) LR/PEO/BT, (c) LP/PEO/BT, (d) LP/PEO, and (e) PAN ACNFs.

average fiber diameters of LB/PEO/BT, LR/PEO/BT, LP/PEO/BT, and LP/PEO ACNFs were 688 ± 80 , 736 ± 122 , 613 ± 63 , and $1012 \pm 74 \text{ nm}$, respectively. The TEM images (Figure 4B), in contrast, show a difference in LB/PEO/BT ACNFs compared to the others. The LB/PEO/BT ACNFs exhibit a highly interconnected porous morphology. The LP/PEO/BT and LP/PEO ACNFs partially show isolated pores, whereas a porous structure is rarely observed in LR/PEO/BT ACNFs. The high-resolution TEM (HR-TEM) image and the selected-area diffraction of the representative ACNFs from lignin exhibit an amorphous carbon structure (Figure S4). In agreement with the HR-TEM findings, the XRD patterns of

Table 2. Surface Elemental Composition of ACNFs Estimated by XPS Analyses, I_D/I_G Ratio Evaluated from Raman Spectra, and Textural Properties of ACNFs

ACNFs	surface elemental composition (%)			I_D/I_G	S_{BET}^a (m^2g^{-1})	V_{total}^b (cm^3g^{-1})	V_{micro}^c (cm^3g^{-1})	V_{meso}^d (cm^3g^{-1})
	C	O	N					
LB/PEO/BT	94.9	5.1	0	0.98	2195	1.41	0.78 (55.3%)	0.63 (44.7%)
LR/PEO/BT	95.4	4.6	0	1.03	1099	0.44	0.41 (93.2%)	0.03 (6.8%)
LP/PEO/BT	95.7	4.3	0	1.05	1120	0.50	0.41 (82.0%)	0.09 (18.0%)
LP/PEO	95.1	4.9	0	1.06	1198	0.50	0.43 (86.0%)	0.07 (14.0%)
PAN	87.0	7.9	5.1	0.98	1071	0.44	0.39 (88.6%)	0.05 (11.4%)

^a S_{BET} : BET surface area. ^b V_{total} : total pore volume obtained from the sum of micropore and mesopore volumes. ^c V_{micro} : t -plot micropore volume.

^d V_{meso} : BJH mesopore volume.

the lignin-based ACNFs show broad (002) and (100) diffraction peaks at $23\text{--}26^\circ$ and $\sim 43^\circ$ 2θ (Figure S5), confirming the amorphous character of turbostratic carbon.

For comparison, ACNFs from PAN, a commercial precursor mostly used for carbon fiber production, were prepared by using the same procedures as for ACNFs from lignins. The fiber diameter of PAN ACNFs (216 ± 21 nm) is much smaller than that of the lignin-based ACNFs (Figure S6 in the Supporting Information).

Figure 4C displays the XPS survey spectra of LB/PEO/BT, LR/PEO/BT, LP/PEO/BT, LP/PEO, and PAN ACNFs. Their surface elemental compositions are summarized in Table 2. The elemental compositions of lignin-based ACNFs are different from those of PAN ACNFs. All ACNFs from lignins contain carbon (C) and oxygen (O) elements on the surface, while PAN ACNFs show the presence of additional nitrogen (N). The carbon content of lignin-based ACNFs ($\sim 94.9\text{--}95.7\%$) is higher than that of the PAN ones ($\sim 87.0\%$). Inversely, the heteroatoms (O and N atoms) on the fiber surface of PAN ACNFs are greater than those (O atoms) on the lignin-based ACNFs.

Raman spectroscopy was used to study the structure and graphitic quality of the ACNFs. The Raman spectra of all samples display two typical broad peaks corresponding to D and G bands of carbon (Figure 4D). The D band at $\sim 1349\text{--}1331\text{ cm}^{-1}$ represents a disordered carbon structure (turbostratic carbon structure), and the G band at $1600\text{--}1594\text{ cm}^{-1}$ represents the graphitic nature of sp^2 carbon. The I_D/I_G are estimated and summarized in Table 2, where I_D and I_G are the intensities at D and G bands, respectively. The I_D/I_G ratio indicated the disorder degree of carbon materials. The I_D/I_G ratio of LB/PEO/BT (0.98) is similar to that of PAN ACNFs (~ 0.98) but lower than that of the others (~ 1.03 to 1.06). The lower the I_D/I_G ratio, the lower the disordering structure of carbon, in other words, the higher the degree of carbon structure ordering. Comparing LP/PEO and LP/PEO/BT, BT incorporated in the lignin-rich nanofibers during electrospinning did not significantly affect elemental compositions and the graphitic structure of the ACNFs.

To probe the conductance of individual ACNFs, we applied cAFM (Figure 5a–e). For these experiments, fibers were deposited onto gold substrates. We used Au substrates to guarantee a good electrical contact to the fibers. Consistent with the SEM results, the AFM topographic images of all lignin-based ACNFs show fiber surfaces with a roughness of $4\text{--}17$ nm determined by a line profile along the fiber at its top side. Applying a positive voltage between the PtIr-coated tip and the Au substrate resulted in an electrical current. We expect that the fibers are orders of magnitude less electrically

conductive than the Au substrate. Thus, upon applying a voltage between the tip and the Au substrate, it exceeds the current limit of the tip apex and breaks the electrical contact. Consequently, on the Au substrate, we detected no current at all. However, a softer sample, where the tip indents deeper, facilitates an electrical contact to the adjacent side wall coating of the tip. In general, bright spots in the current images indicate the areas of electrical currents. In particular, some pixels appear white, indicating that the local current exceeds the current limit of the electrical amplifier (>120 nA). A higher current was typically measured at the edge of the fiber. We attributed this higher current to an increase in the contact area of the tip and the fiber. Here, not the tip is in contact with the fiber but a larger portion of the tip's side wall.

At a voltage of 0.5 V, the current image of LB/PEO/BT ACNFs presents a uniform distribution of current signals throughout the fiber surface (Figure 5A-line a). For all other fibers, a higher current is located mainly at their edges (Figure 5A lines b–e). The uniform distribution of the current signals on the surface of LB/PEO/BT ACNFs indicated continuous conductive phases across the fibers. In order to compare local electrical conductance of the individual single nanofibers, we determined the number of the current signals >0.1 nA ($N_{I>0.1\text{nA}}$) and divided this number by the area of the fibers (A). The ratios of $N_{I>0.1\text{nA}}/A$ were calculated from three different fibers of each sample (Figure 5B). The cAFM results unveiled that the ratio of $N_{I>0.1\text{nA}}/A$ for LB/PEO/BT was much higher than that of PAN, LR/PEO/BT, LP/PEO, and LP/PEO/BT, suggesting the highest electrical conductivity of LB/PEO/BT. This conclusion was consistent with our Raman spectroscopic data where the good electrical properties of LB/PEO/BT were associated with a high content of ordered graphitic carbon (low I_D/I_G ratio). Note that although PAN and LB/PEO/BT ACNFs exhibited similar I_D/I_G ratios, PAN ACNFs had lower electrical conductivity than LB/PEO/BT ACNFs. This might be due to the high number of heteroatoms in PAN ACNFs.

The N_2 adsorption–desorption isotherms of the ACNFs at 77 K are presented in Figure 6a. LR/PEO/BT, LP/PEO/BT, LP/PEO, and PAN ACNFs exhibit a typical type I isotherm with a steep uptake of N_2 at low relative pressure ($P/P_0 < 0.1$) and a plateau at $P/P_0 \sim 0.1\text{--}1.0$, similar to the activated carbon from lignin powders (Figure S3 in the Supporting Information). As demonstrated in Figure 6b, the pore sizes of LR/PEO/BT, LP/PEO/BT, LP/PEO, and PAN ACNFs are mainly distributed in the range of $0.5\text{--}0.8$ nm with the maximum pore size at ~ 0.6 nm. For LB/PEO/BT, the N_2 adsorption isotherm increased sharply at low relative pressure ($P/P_0 < 0.1$) and then continued to increase with a lower slope

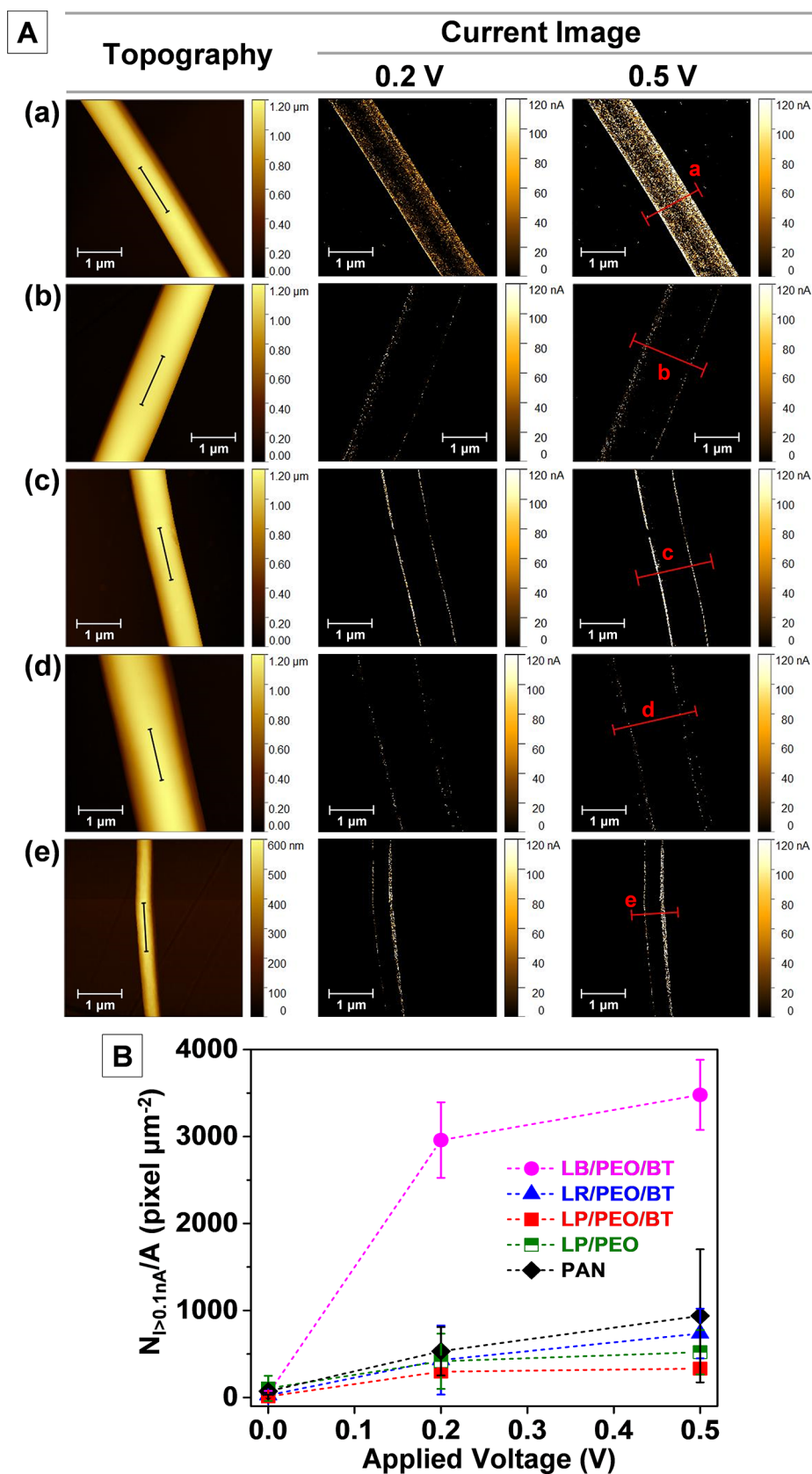


Figure 5. (A) Topography and current images of (a) LB/PEO/BT, (b) LR/PEO/BT, (c) LP/PEO/BT, (d) LP/PEO, and (e) PAN ACNFs at voltages of 0.2 and 0.5 V and (B) plot of $N_{I>0.1 \text{ nA}}/A$ as a function of applied voltage.

at medium and high relative pressure ($P/P_0 \sim 0.2\text{--}1.0$). This shape indicated a coexistence of micropores, mesopores,

macropores, and external surface.^{62,63} Micropores with pore sizes <2 nm and mesopores with pore sizes in the range of 2–

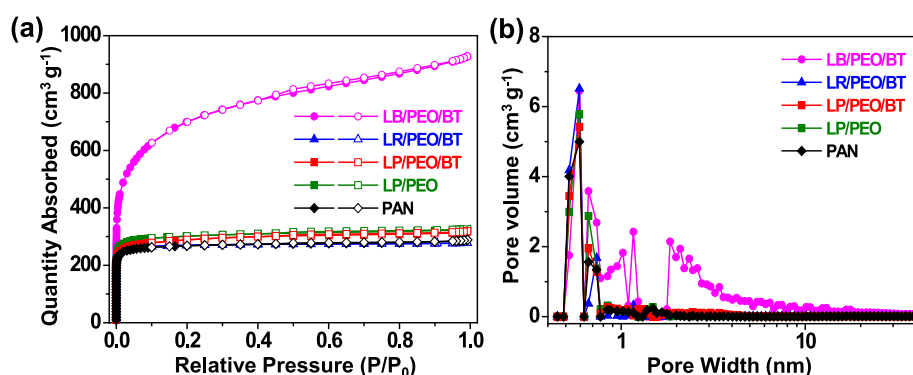


Figure 6. (a) N₂ adsorption–desorption isotherms at 77 K and (b) pore size distribution of LB/PEO/BT, LR/PEO/BT, LP/PEO/BT, LP/PEO, and PAN ACNFs.

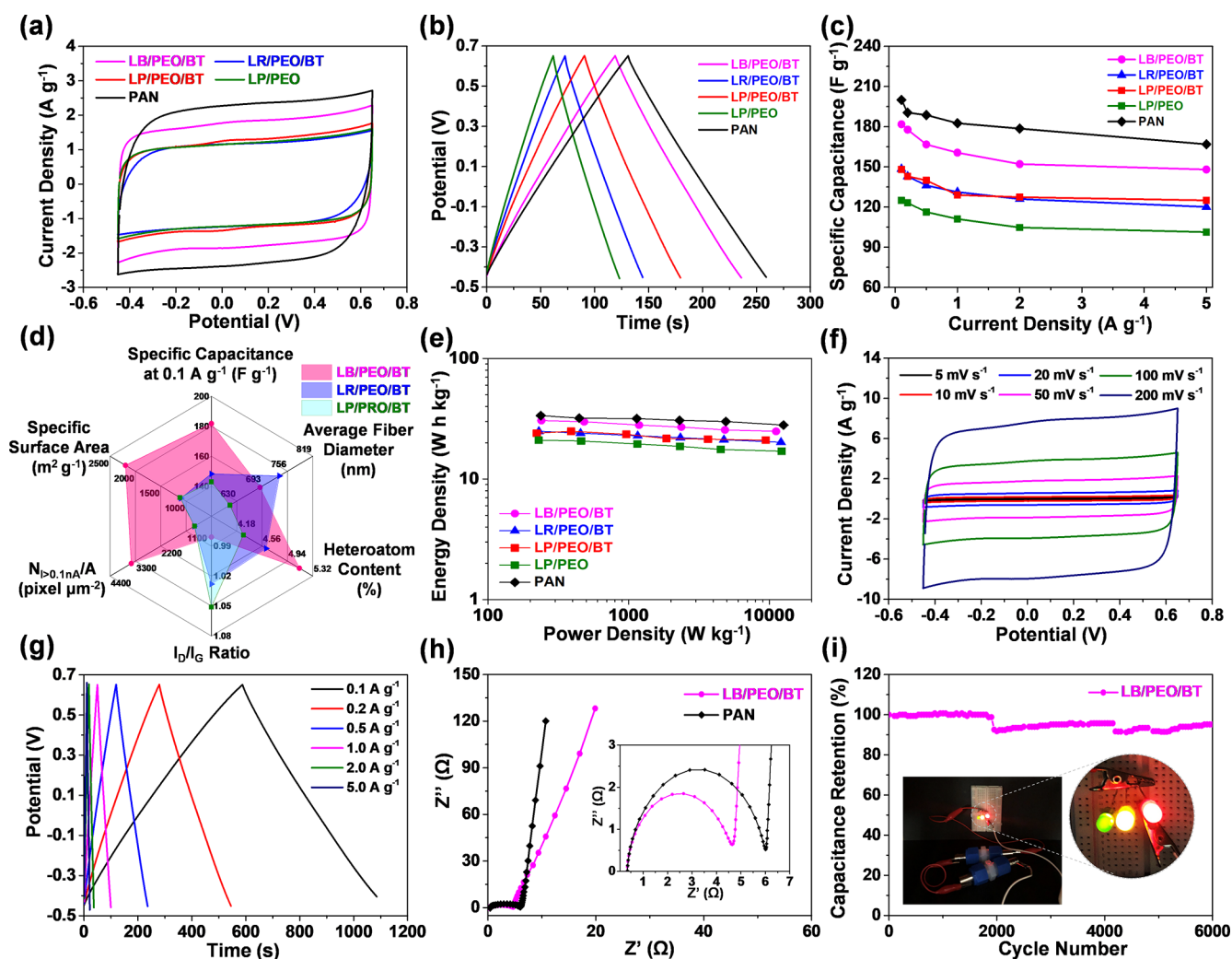


Figure 7. (a) CV curves of ACNFs from lignins and PAN at 50 mV s⁻¹ in a 1.0 M H₂SO₄ electrolyte, (b) GCD curves of all ACNF samples at 0.5 A g⁻¹, (c) plot of specific capacitance at different current densities, (d) radar plot of specific capacitance at a current density of 0.1 A g⁻¹, average fiber diameter, heteroatom content, I_D/I_G ratio, N_{D>0.1nm}/A, and specific surface area for LB/PEO/BT, LR/PEO/BT, and LP/PEO/BT ACNFs, (e) Ragone plot, (f) CV curves of LB/PEO/BT ACNFs at different scan rates, (g) GCD curves of LB/PEO/BT ACNFs at different current densities, (h) Nyquist plot of LB/PEO/BT ACNFs and PAN ACNFs, and (i) cycle stability of LB/PEO/BT ACNFs at a current density of 0.1 A g⁻¹. The inset is photograph of red, yellow, and green light-emitting diodes power-supplied by a supercapacitor using LB/PEO/BT ACNFs as electrodes.

20 nm were indeed observed in the LB/PEO/BT ACNFs. The specific surface area was calculated based on the BET method. The mesopore volume and micropore volume were estimated using Barrett–Joyner–Halenda (BJH) and t-plot methods,

respectively. The resulting specific surface area and pore properties are summarized in Table 2. The specific surface areas of the lignin-based ACNFs (1099–2195 m² g⁻¹) are much higher than those of the activated carbon powders from

lignins ($452\text{--}590\text{ m}^2\text{ g}^{-1}$). Therefore, the preparation of activated carbons in the form of nanofibers improved the specific surface areas. The LB/PEO/BT ACNFs from high-purity lignin have the highest specific surface area ($2195\text{ m}^2\text{ g}^{-1}$) and total pore volume ($1.41\text{ cm}^3\text{ g}^{-1}$). The specific surface area of LR/PEO/BT, LP/PEO/BT, and LP/PEO ACNFs is in the range of $1099\text{--}1198\text{ m}^2\text{ g}^{-1}$, similar to that of PAN ACNFs.

Unlike our findings, Han et al. reported that the activated carbon from milled wood residues with a low lignin content exhibited a higher specific surface area than the one with a high lignin content due to chemical inertness of the aromatic-rich structure of lignin.⁸ Therefore, this might be not only due to purity of lignin contributed to the development of surface area and carbon porosity but also due to other factors such as the S/G ratio,^{8,36,38} PDI,⁸ and fiber diameter.⁶³

The total pore volume of lignin-based ACNFs was in the order of LB/PEO/BT > LP/PEO/BT $\sim$ LP/PEO > LR/PEO/BT. The micropore volumes of LB/PEO/BT, LR/PEO/BT, LP/PEO/BT, LP/PEO, and PAN ACNFs are 0.78, 0.41, 0.41, 0.43, and $0.39\text{ cm}^3\text{ g}^{-1}$, respectively, which are approximately 55.3, 93.2, 82.0, 86.0, and 88.6% of the total pore volumes. Comparing LP/PEO/BT and LP/PEO, it seems likely that the addition of BT for 10 wt % had no significant effect on the surface area and total pore volume but slightly increased the mesoporous volume fraction of the materials from 14 to 18%.

Electrochemical Performance. The electrochemical performance of the lignin-based ACNFs from different biomass feedstock was investigated for supercapacitor application. The CV, GCD, and EIS profiles of PAN ACNFs were also analyzed and reported as a reference material. The electrochemical measurement was performed in a two-electrode cell using $1.0\text{ M H}_2\text{SO}_4$ as an electrolyte. The CV profiles of all samples at a scan rate of 50 mV s^{-1} and the GCD curves at a current density of 0.5 A g^{-1} are presented in Figure 7a,b, respectively. Similar to PAN ACNFs, all lignin-based ACNFs exhibit nearly rectangular shapes of the CV curves and symmetric triangular shapes of the GCD curves, indicative of dominant EDLC behavior with superior charge/discharge reversibility. In the potential range of -0.45 to $+0.65\text{ V}$, the CV curve of PAN ACNFs displays a larger current density than that of lignin-based ACNFs. For the ACNFs from lignins, LB/PEO/BT exhibits a larger CV curve area than the other three samples, suggesting a better charge storage capability. Likewise, LB/PEO/BT shows a longer charge/discharge time than LP/PEO/BT, LR/PEO/BT, and LP/PEO, respectively.

The specific capacitance (C_s) calculated from discharge curves at various current densities is plotted in Figure 7c. At a current density of 0.1 A g^{-1} , the C_s of LB/PEO/BT, LR/PEO/BT, LP/PEO/BT, and LP/PEO was 182, 148, 143, and 125 F g^{-1} , respectively, which were lower than that of the PAN ACNFs (200 F g^{-1}). The higher C_s of PAN ACNFs compared to the lignin-based ACNFs was possibly due to the small diameter of the fibers with nitrogen functionalization. As revealed by Candelaria et al., the presence of N atoms improved the wettability of the carbon, resulting in enhancement of charge storage performance.⁶⁴

Similar to that of PAN ACNFs, the C_s of lignin-based ACNFs decreased with increasing current density, but it could retain more than 80% capacitance at a current density as high as 5.0 A g^{-1} . In the entire current density ranging from 0.1 to 5.0 A g^{-1} , LP/PEO/BT had higher specific capacitance than LP/PEO. This revealed the role of BT in improving the

specific capacitance of the ACNFs from lignin precursors. Comparing the ACNFs from lignins with 10 wt % BT, the specific capacitance of LB/PEO/BT was higher than that of LR/PEO/BT and LP/PEO/BT. Figure 7d shows a radar plot summarizing various properties of the LB/PEO/BT, LR/PEO/BT, and LP/PEO/BT ACNFs including specific capacitance from electrochemical analysis, fiber diameter from SEM images, heteroatom content from XPS, I_D/I_G ratio from Raman spectroscopy, $N_{I>0.1\text{ nA}}/A$ from cAFM, and specific surface area from BET data. It was revealed that the highest specific capacitance of LB/PEO/BT was possibly attributed to many factors, that is, small fiber diameter, high content of heteroatom, high graphitic carbon structure, high conductivity, and high specific surface area. The Ragone plot demonstrating energy density and power density of all ACNFs is shown in Figure 7e. Among all lignin-based ACNFs, LB/PEO/BT ACNFs exhibited the highest energy density of 31 W h kg^{-1} at a power density of 240 W kg^{-1} , whereas PAN ACNFs provided a maximum energy density of 34 W h kg^{-1} at a power density of 238 W kg^{-1} . Although LB/PEO/BT had lower specific capacitance and energy density than PAN ACNFs, it showed a great potential for electrochemical energy storage. As illustrated in Figure 7f, almost rectangular CV curves of the LB/PEO/BT ACNFs are maintained even upon increasing the scan rates up to 200 mV s^{-1} , revealing their excellent rate capability and reversibility.⁶⁵ The LB/PEO/BT ACNFs show symmetric linear GCD curves with an insignificant IR drop at different current densities (Figure 7g), suggesting excellent capacitive performance with low cell resistance and good charge/discharge reversibility.^{66,67}

The Nyquist plot of lignin-based ACNFs and PAN ACNFs shows a similar pattern with a semicircle in the high-frequency region and a spike nearly parallel to the imaginary axis in the low-frequency region (Figure 7h). At the highest frequency, the intercept on the real axis reflects the bulk resistance (R_{bulk}), that is, the sum of resistance of the bulk electrolyte, active material, and current collector.⁶⁸ The R_{bulk} values of both LB/PEO/BT and PAN ACNFs were as small as $0.3\text{ }\Omega$, indicating that the dominant electrical effect was from the electrolyte. The semicircle represents an interfacial impedance that can be associated with ion transport processes between the electrode and the electrolyte described as the charge-transfer resistance and the double-layer capacitance^{68,69} as well as the contact impedance between the electrode and the current collector.⁶⁹ Here, the diameter of a semicircle was simply estimated to be the value of the interfacial resistance (R_{inf}).⁷⁰ The R_{inf} value of LB/PEO/BT ACNFs ($4.4\text{ }\Omega$) was less than that of PAN ACNFs ($5.7\text{ }\Omega$), suggesting a slightly lower internal resistance of the capacitor cell.⁷⁰ At low frequency, a single vertical straight line with no 45° slope confirmed EDLC behavior of the electrodes. The nearly vertical lines at low frequency suggested that the electrodes behaved close to an ideal EDLC. The noticeable nonvertical spike in LB/PEO/BT could be inherent to the relatively large pore resistance than that of PAN ACNFs, probably due to the difference in the pore structure.⁷⁰

Figure 7i illustrates the cycling performance of the LB/PEO/BT electrodes. The LB/PEO/BT ACNFs exhibited good cyclability with a capacitance retention as high as 95.7% after 6000 cycles of charge/discharge testing at a current density of 0.1 A g^{-1} . The slight reduction in retention capacitance was due to the damage of the ACNFs after cycle testing (Figure S7). The inset of Figure 7i demonstrates the application of LB/

PEO/BT ACNFs as binder-free electrodes for supercapacitors. After charging, two symmetric cells with ~ 40 mg of LB/PEO/BT as electrodes and 1.0 M H_2SO_4 as an electrolyte could deliver power to light up the red, yellow, and green diodes.

CONCLUSIONS

The lignin-rich nanofibers from LB, LR, and LP were successfully prepared via electrospinning with solid content of lignins up to 99%. To improve the spinnability of lignin, PEO was added. The difference in sources of lignin had no significant effect on fiber spinnability in the presence of PEO. The source, however, greatly influenced the thermal stability of the nanofibers. The fibrous structures of LB/PEO and LR/PEO were not stable at high temperature. In contrast, the LP/PEO nanofibers could be stabilized and carbonized at 900 °C while preserving their fibrous morphology. In this study, BT was used as a cross-linker to improve the structural stability of the lignin-rich nanofibers when heat-treated. Herein, the ACNF sheets were successfully produced from LB/PEO, LR/PEO, and LP/PEO nanofibers with 10 wt % of BT via thermal stabilization, carbonization, and activation. BET analysis revealed that the specific surface area of the ACNFs from lignin-rich materials was ~ 2 – 4 times higher than the activated carbon powder from pure lignins. The addition of BT not only prevented softening of lignins during oxidative thermostabilization but also increased the specific capacitance of the ACNFs. The composition, surface area and porosity, graphitic carbon, electrical property, and electrochemical performance of the ACNFs were different depending on the source of lignin. Among all lignin-based ACNFs, due to small size of the fibers, a high specific surface area, high contents of graphitic carbon and heteroatoms, and high conductivity, LB/PEO/BT exhibited the highest specific capacitance of 182 F g^{-1} (at 0.1 A g^{-1} in 1.0 M H_2SO_4) which had 91% capability of the ACNFs from PAN. The LB/PEO/BT ACNFs possessed an excellent performance for supercapacitor applications with their high energy density (31 W h kg^{-1} at a power density of 240 W kg^{-1}) and outstanding cycling stability (95.7% capacitance retention after 6000 cycles). The results demonstrate that the LB/PEO/BT ACNFs can be applied as binder-free electrodes for high-performance supercapacitors.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsanm.1c02637>.

Composition, ash content, weight-average molecular weight (M_w), number-average molecular weight (M_n), and polydispersity index (PDI) of organosolv lignins from different sources; 2D-HSQC NMR spectra in aromatic region of LB, LR, and LP showing correlation signals of $S_{2,6}$, $S'_{2,6}$, and G_2 substructures; SEM images of LB, LR, and LP powders before and after stabilization, carbonization, and activation; N_2 adsorption–desorption isotherms at 77 K and pore size distribution of activated carbon powders from LB, LR, and LP; Textural properties of activated carbon powder from lignins; HR-TEM image and selected-area diffraction of representative LB/PEO/BT ACNFs; XRD patterns of LB/PEO/BT, LR/PEO/BT, LP/PEO/BT, and LP/PEO ACNFs; SEM image of ACNFs from PAN; and

SEM image of LB/PEO/BT nanofiber sheets after 6000 cycles of the charge/discharge test (PDF)

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Notes

The authors declare no competing financial interest.

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